

Ho Nghia Khang

AI Engineer / Full-stack Developer

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Professional Summary

AI Engineer and Full-stack Developer with 5+ years of experience turning computer vision, backend systems, and production AI workflows into reliable business products. Strong in Python, FastAPI, LLM/RAG, Docker, Kafka, AWS, CI/CD, monitoring, and GPU-based image/video processing. Currently focused on building safe, practical AI agents for small and medium-sized businesses, while helping non-technical teams understand, adopt, and apply AI effectively.

Technical Skills

Languages	Python, SQL, Shell Scripting
AI / ML	Computer Vision, Image Processing, Object Detection, Segmentation, OCR, LLMs, Generative AI, RAG, PyTorch, Keras, OpenCV, Scikit-learn, NumPy, Pandas
AI Agent Tools	Cursor, Claude, Codex, Qwen CLI, OpenCode, Antigravity, Google AI Studio, NotebookLM, MCPJam
Backend	FastAPI, REST APIs, Celery, Kafka, asynchronous processing, API design, microservices
Frontend	ReactJS, NextJS, Streamlit
DevOps / MLOps	Docker, Portainer, GitLab CI/CD, GitHub Actions, Ansible, Linux, automated deployment, rollback strategies
Monitoring	Prometheus, Grafana, Sentry, alerting, production observability
Databases	MongoDB, PostgreSQL, PGVector, Redis, MySQL
Cloud / Tools	AWS EC2, S3, Lambda, Batch, RDS, SES, VPC, Git, GitLab, SonarQube, Postman, VSCode, PyCharm, Excalidraw, Obsidian

Professional Experience

AI Engineer <i>Mynavitech</i>	Dec 2025 - Present Danang, Vietnam
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- Developed applied AI solutions across machine learning, computer vision, backend integration, workflow automation, and deployment readiness.
- Researched and built AI agents for internal operations and product-oriented use cases.
- Tracked emerging AI models and event-driven AI patterns to identify integration opportunities that improve organizational productivity.

Backend Developer <i>Rikai Technology</i>	Sep 2025 - Nov 2025 Danang, Vietnam
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- Automated server setup for AI training and inference workloads using Ansible.
- Built a resource allocation system that partitions high-performance GPU servers into smaller rentable units for individual users.
- Designed APIs for the GPU resource management and rental platform.

Python Developer <i>PiconTechnology</i>	Feb 2025 - Jun 2025 Danang, Vietnam
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- Developed backend features and proof-of-concept solutions in collaboration with business analysts and frontend engineers.
- Worked with PostgreSQL for data modeling, CRUD operations, and backend integration.
- Integrated AWS services including S3, Lambda, and Batch to support scalable, event-driven backend workflows.
- Gained practical experience with project structure, development workflow, and cross-functional delivery processes.

AI Engineer / DevOps Engineer <i>FTECH CO., LTD</i>	Nov 2022 - Feb 2025 Danang, Vietnam
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- Designed and implemented AI modules based on client requirements, focusing on image processing, computer vision, and resource optimization.
- Designed, deployed, and maintained microservices for long-running AI and video-processing jobs using FastAPI, Kafka, MongoDB, and Docker.
- Built asynchronous API workflows that return immediate responses while background workloads run for extended CPU/GPU processing periods.

- Implemented monitoring and alerting with Prometheus, Grafana, and Sentry to improve production visibility and incident response.
- Established CI/CD pipelines for linting, testing, deployment, release automation, and rollback across production services.
- Guided junior team members on AI development, DevOps practices, Git, Docker, and performance reporting.

AI Intern

FTECH CO., LTD

Dec 2021 - Nov 2022

Danang, Vietnam

- Researched and developed computer vision tasks including object detection, image classification, segmentation, 3D modeling, and OCR.
- Focused on semantic segmentation tasks such as hair segmentation, face parsing, and human parsing.
- Improved research skills through paper reading, experiment tracking, and technical report writing.

Selected Projects

API Services for Long-Running Tasks

FTECH

Sep 2023 - Jan 2025

- Developed and maintained a microservice-based API system for CPU/GPU-intensive tasks that may require hours of processing time.
- Designed immediate-response APIs with asynchronous processing to improve user experience, fault tolerance, and system scalability.
- Applied Kafka-based workload distribution and separated services to scale processing capacity.
- Built monitoring and alerting with Prometheus, Grafana, and Sentry for production performance tracking.
- Designed caching and request-control mechanisms to reduce repeated workloads and protect services from excessive duplicate requests.
- Implemented CI/CD automation for linting, testing, deployment, and rollback.

Automatic Image and Video Merging with Green Screen

FTECH

Sep 2023 - Jan 2025

- Led development of a module that automatically merges images and videos into predefined templates, including videos with green-screen backgrounds.
- Integrated deep learning models and image-processing logic to improve merging precision and output consistency.
- Set up production alerting to detect processing errors and improve reliability.
- Optimized processing time and resource usage for production workloads.
- Mentored a fresher on software development processes, Git, Docker, metrics, and technical reporting.

Soccer Match Analysis

Freelance Project on Upwork

Apr 2025 - Present

- Engineered a cross-platform desktop application using computer vision to support football clubs with tactical and performance analysis.
- Developed a video-stitching pipeline for two wide-angle camera feeds, including camera calibration and lens distortion correction.
- Built an intelligent auto-focus module using object detection and clustering to simulate camera panning toward active player zones.
- Independently managed client communication, technical design, implementation, packaging, and delivery.
- Achieved high client satisfaction and completed the first USD 1,000 milestone.
- Demo: youtube.com/watch?v=ZNacjRtL9JQ

LectureHub

Personal Project

Jun 2025 - Present

- Built a full-stack web platform that helps users look up solutions for competitive programming problems from multiple online judges.
- Developed a structured web application, implemented CI/CD, and deployed the system on AWS.
- Designed an advanced Q&A chatbot PoC using Retrieval-Augmented Generation with a vector database to answer problem-specific queries with reduced hallucination.
- Website: lecturehub.tech Repository: github.com/LectureHubTeam/lecturehub-chatbox

CodeMath Solver

Personal Project

Apr 2025 - Present

- Engineered an autonomous problem-solving agent for programming challenge platforms using LLM prompting and browser automation.
- Designed a multi-stage prompt engineering workflow for solution generation, code formatting, and stable automated submission.
- Built a resilient Selenium-based automation pipeline with exception handling for long-running operation.
- Achieved the No. 1 rank on codemath.vn within 24 hours of deployment.
- Repository: github.com/LectureHubTeam/codemath-solver

License Plate Recognition

Jul 2022 - Sep 2022

Freelance Project

- Developed and deployed an automated license plate recognition system for a factory environment.
- Achieved 97% accuracy on a private real-world test dataset.
- Built a Docker image and API to simplify deployment.
- Repository: github.com/KudoKhang/LPR

Education

Danang University of Science and Technology

2026 - Present

Master's Program in Mechatronics Engineering

Danang University of Science and Technology

2017 - 2022

Programme de Formation d'Ingenieurs d'Excellence au Vietnam (PFIEV)

Bachelor of Engineering in Automation

Publications & Intellectual Property

- Automated Facial Blemish Removal Process Leveraging Image Processing and Deep Neural Networks. Intellectual Property Application, Utility Solution No. 4976/SHTT-SC.IP, dated Apr 3, 2026; issued by the Ministry of Science and Technology - Intellectual Property Office of Vietnam.
- Automated Panoramic Video Generation and Moving Object Tracking from Dual Fixed Cameras Using Image Stitching and Electronic Pan-Tilt-Zoom Techniques. Intellectual Property Application, Application No. 7314/SHTT-SC.IP, dated Oct 27, 2025; issued by the Ministry of Science and Technology - Intellectual Property Office of Vietnam; status: pending examination.
- Dang Phuoc Vinh, Ho Nghia Khang, Doan Le Anh. *Using Convolutional Neural Networks for License Plate Recognition at Woodchip Factories*. Proceedings of the 7th International Conference on Advanced Technologies in Green Buildings (ATiGB 2022), ISBN: 978-604-80-7240-7, pp. 1–6.

Additional Experience

Programming Educator & Mentor

May 2024 - Present

Da Nang

- Teach Scratch and Python fundamentals, basic algorithms, problem solving, and computational thinking through age-appropriate coding exercises.
- Planned learning roadmaps, designed lesson materials, and adapted coding exercises for primary and secondary students with different skill levels.
- Managed classroom engagement and explained difficult programming concepts in simple, visual, and practical ways for young learners.
- Website: scratchzone.tech

References

Assoc. Prof. Dang Phuoc Vinh, Lecturer at Danang University of Science and Technology.

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